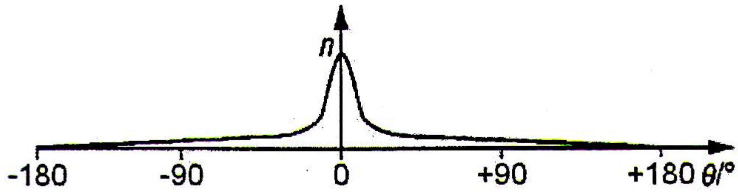
α-particles are projected towards a thin gold foil placed in an evacuated chamber(to prevent air molecules from slowing the α-particles). The distribution of α-particles after passing through the gold foil can be measured using a movable detector mounted on a circular rail surrounding the set up. Most of the α-particles are found top pass through undeflected, with a small number being deflected, and an even smaller number (fewer than $\frac{1}{8000}$) deflected by more than $90°$. The distribution will have a sharp peak at $0°$ measured from the straight-through axis, and falls sharply to both sides of the centre.



This is the Geiger and Marsden's experiment , which confirms Rutherford's model of the atom that:

- The atom is mostly empty space with a small nucleus at its centre – this accounts for the fact that most of the α-particles pass straight through this empty space undeflected;
- The nucleus is massive – this accounts for its ability to deflect high-speed α-particles coming close to it;
- The nucleus is positively-charged – this accounts for its ability to deflect backward high-speed α-particles coming head on towards it, resulting in a detectable count rate beyond $\pm 90°$ from the straight-through axis.

Note: A surprisingly large number of candidates did not realise that the experiment to be described involves α-particles and gold film/foil. There was confusion between the terms 'atoms' and 'nuclei'. Many thought that the particles travelled without deflection through the nucleus. Others thought that the 'empty space' is between atoms. Candidates should be advised that, when referring to a 'small' nucleus, then they should make a comparison with the atom.

(b) (i) $x = 138$
$y = 40$